



Habib University - City Campus
Course Title: CS/CE 224/272 Object Oriented Programming and Design
Methodologies
Section : L4
Instructor's names: Nadia Nasir

Templates in C++

Activity for attendance: 15th Nov (slot : 3:30 to 4:45)

Question: Define the Employee class with private members the name (e.g., name) and the salary (e.g., sal) of the employee. From the Employee class derive with public access the Manager class with private member the name of the department (e.g., dpt). Each class must contain a public show() function to display the values of its members. Next, define the template Person<T> class, where T can be a pointer to an Employee or Manager type. The class should contain a private member of type vector (e.g., v) in which the T objects are stored. To store each object you should define an appropriate add() function. Also, the class should contain a show() function which calls the show() of the objects contained in the vector object to display their elements. Add appropriate functions in the classes so that the following program works.

```
int main()
{
    Employee emp1("First", 1), emp2("Sec", 2); /* name should become equal
to the value of the first argument, and sal with that of the second. */
    Manager mgr1("Prog", "Third", 3), mgr2("Net", "Four", 4); /* dpt should
become equal to the value of the first argument and the constructor of the
Employee class should be called. */
    Person<Employee*> per1;
    per1.add(&emp1); // The argument should be stored in per1.v
    per1.add(&emp2);
    per1.show(); /* The program should display the data of the objects,
that is, First 1 and Sec 2. */
    Person<Manager*> per2;
    per2.add(&mgr1); // The argument should be stored in per2.v
    per2.add(&mgr2);
    per2.show(); /* The program should display the data of the objects,
that is, Third 3 Prog and Four 4 Net. */
    return 0;
}
```